



**HP407 – EVIDENCE REVIEW AND
SYNTHESIS FOR DECISION MAKING**

SUMMATIVE ASSESSMENT 2022/23

CANDIDATE NUMBER:	<table><tr><td>5</td><td>1</td><td>6</td><td>0</td><td>0</td></tr></table>	5	1	6	0	0
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ASSESSMENT TITLE	Efficacy of caregiver interventions in low- and middle-income countries: a meta-analysis and systematic review					
WORD COUNT						

Please tick this box if you would be happy for your assessment to be shared with future cohorts: ☒

Efficacy of caregiver interventions in low- and middle-income countries (LMICs): a meta-analysis and systematic review

ABSTRACT

Background

More than two thirds of dementia patients live in LMICs, where the family is often the main caregiver and the support for caregivers are scarce. Distinct socioeconomic status and clinical profiles of dementia patients and their carers stresses the need to analyse the effectiveness of caregiver interventions in the LMIC settings.

Objective

To evaluate the effects of training and support programmes for caregivers of PwD on caregiver burden and quality of life in LMIC settings.

Design

Five reviewers screened title and abstracts. The risk of bias and data of each study was assessed by at least two reviewers, while disagreement was solved through group discussion among the reviewers.

Data sources

We further included 7 studies that focuses on support for caregivers published from 2019 to 2020 and updated the previous meta-analysis of 733 studies in LMICs, where 26 studies were found to have a significant proportion of caregiver interventions. The 34 studies, based on the type of measurements and data availability, are further curated. Finally, 14 and 12 studies are included for the meta-analysis of caregiver burden and caregiver quality of life, respectively.

Eligibility criteria

Only placebo-controlled RCTs in English and Chinese were included. Included studies should be published between 2019 and 2020. The intervention under study was support or training programmes for caregivers and measured any of caregiver burden, care mental health and caregiver behaviour.

Results

We included 33 studies (n=)

Conclusion

HP407

Candidate No. 51600

TABLE OF CONTENT

ABBREVIATIONS

- BAI Beck Anxiety Inventory
- BDI Beck Depression Inventory
- CBI Community-based intervention
- CES-D 10 10-item Center for Epidemiologic Studies Depression Scale
- CI Confidential interval
- CSI Caregiver Strain Index
- GHQ-28 28-item General Health Questionnaire
- HIC High-income country
- IADL Brody's Instrumental Activities of Daily Living Scale
- LMIC Low- and middle-income country
- NPI-D Score of distress of Neuropsychiatric Inventory
- NPI-S Score of severity of Neuropsychiatric Inventory
- PHQ-4 Patient Health Questionnaire-4
- PICOS People, Intervention, Comparison, Outcome, Study design
- PRISMA Preferred Reporting Items for Systematic Reviews and Meta-Analyses
- PwD People with dementia
- QoL Quality of life
- RCT Randomized controlled trial
- RoB Risk of bias
- SMD Standardised mean difference
- SD Standardised deviation
- SE Standardised error
- SF36 36-Item Short Form Survey
- SQ Spielberger's questionnaire
- ZBI Zarit Burden Interview

INTRODUCTION

Caregivers play a crucial role in providing support to individuals with dementia, but they often face significant challenges, especially high levels of stress and burden, making them the "invisible second patients" of dementia. Training and support interventions for caregivers have been proposed as a potential strategy to address these challenges and improve the quality of care for PwD.¹ However, the effectiveness of these interventions varies according to the accessibility of healthcare resources and the caregivers' knowledge and awareness of dementia care, which can affect their willingness to seek support.²

In LMICs, where more than two thirds of individuals with dementia live,³ caregivers are often the main source of support, but support for caregivers is scarce.^{4,5} Previous systematic reviews and meta-analyses have identified several types of interventions, such as psychoeducational interventions and multicomponent interventions, that can significantly alleviate caregiver burden.⁶ However, these interventions are not all implemented and rarely validated in LMIC settings. Furthermore, a recent meta-analysis that includes studies from both LMICs and HICs suggests mixed evidence for coordinated community interventions.⁷

Given the unique clinical manifestations of dementia and inequalities in health resources and awareness of caregiver support in LMICs,⁸ it is necessary to evaluate the effectiveness of caregiver interventions specifically in LMIC settings. This systematic review and meta-analysis aims to evaluate the effectiveness of training and support interventions for caregivers of people with dementia in LMICs. In this paper, we will also explore the potential moderators of the interventions, thus providing insights into effective caregiver support in dementia care.

METHODS

Identification and selection of studies

In addition to 733 records retrieved from the previous meta-analysis that we aim to update,¹²¹²⁵ a database search of Ovid Medline, Ovid Embase, Ovid Global Health,

Commented [MOU1]: The **methods** should section should provide an overview of the methodological choices and decisions and their justifications.

Commented [MOU2]: *Eligibility criteria:* Specify study characteristics (such as PICO, length of follow-up) and report characteristics (such as years considered, language, publication status) used as criteria for eligibility, giving rationale. Across these sub-headings, the methods section should include information on all of the following elements (These are taken directly from the PRISMA statement and checklist):
Information sources: Describe all information sources (such as databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched
Study selection: State the process for selecting studies (that is, screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis)

Ovid PsycINFO, EBSCO CINHL was performed to identify relevant studies published during 2019-2020 by a librarian, with the search strategy described in Appendix 1. The selection process is depicted in Figure 1. Rayyan was used for automated keyword extracting and highlighting and duplicate detection and removal.⁹ Decisions were made through the consensus of all four reviewers. Inclusion criteria include:

- 1) Publication: RCT published in English or Chinese
- 2) Population: PwD and their caregivers in LMICs
- 3) Intervention: support or training for caregiver
- 4) Comparators: No intervention as a control
- 5) Outcomes: measuring the effect of caregiver support on patient and caregiver health

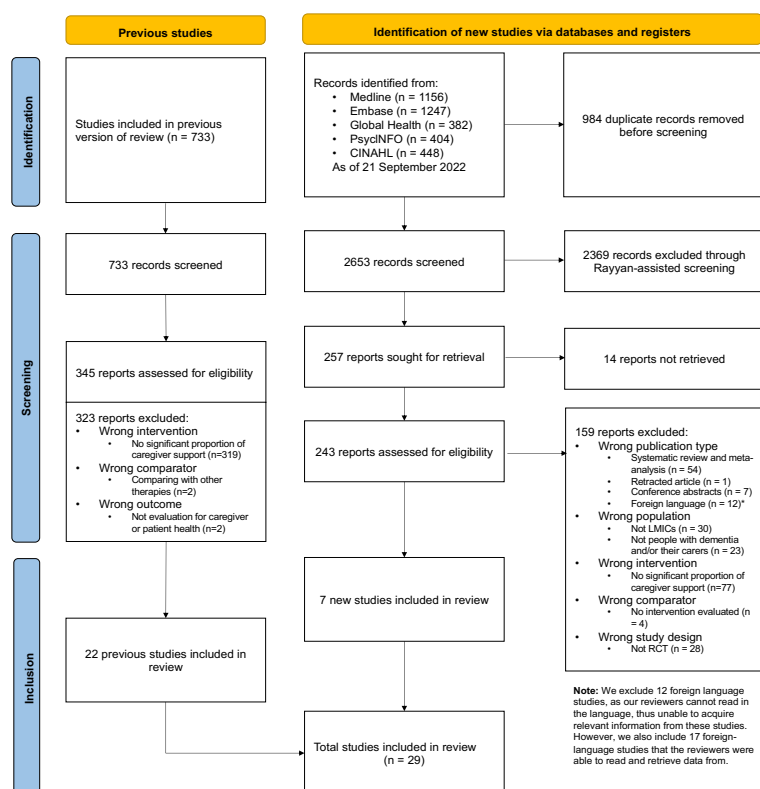


Figure 1. Flow chart of paper screening (PRISMA)

Only RCTs were included because it is considered the most reliable evidence on the effectiveness of an intervention due to its scientifically rigorous methods, including but not limited randomisation, minimising the risk of the confounding factors affecting the results.¹⁰ The same criteria were applied to selection of the records from the 2019 meta-analysis. Ultimately, 7 studies from the 2022 search and 22 studies from the 2019 meta-analysis were included.

Data extraction

5 reviewers together identified the outcome measures to be extracted the data from the 7 newly included studies, with at least two reviewers recorded data items. One reviewer extracted the data from 26 previous studies using R,¹¹ with the detailed procedures described in Appendix 4. Outcome measures related to the perceived burden, health and quality of life of either dementia patients or their caregivers were collected, while those measuring the knowledge and attitude of the caregiver were discarded as they were not directly linked to the patient or caregiver health.

One study measured the subjective well-being and the positive experience of caregivers,¹² and another measured caregiver self-efficacy.¹³ These outcome measures were not well-studied but closely associated with QoL.¹⁴ Two studies only measured the mental and physical health perspectives of QoL.^{15,16} Thus, although these data were collected, but they were insufficient to be meta-analysed. The collected outcome measures for caregivers and care recipients were listed in Table 1 and Table 2, respectively.

Table 1. Summary of caregiver outcome measures¹

	<i>Perceived burden</i>	<i>Quality of life</i>	<i>Depression</i>	<i>Anxiety</i>	<i>Neuropsychiatric symptoms</i>
<i>Amit 2008</i>	ZBI				
<i>Chen 2020</i>					
<i>Ghaffari 2019</i>			GHQ-28	GHQ-28	GHQ-28
<i>Govindakumari 2020</i>	Unspecified				
<i>Guerra 2011</i>	ZBI	WHO-QoL-Bref			NPI-D NPI-S SRQ-20 PHQ-4
<i>Hinton 2020</i>	ZBI				

¹ Please refer to the Abbreviations section for measurement methods used.

JIANG 2012		WHO-QoL-Bref			
Lin 2017	CBI				
Lök 2017	ZBI				
Mahdavi 2017	CSI				
Mercedeh 2014	CBI	SF-36	Ham-D		
Orawan 2018					
Pahlavanzadeh 2010	ZBI				
Pan 2019			CES-D		
Salamizadeh 2017					
Shata 2017	ZBI				
Söylemez 2016	CBI	WHO-QoL-Bref			
SU 2012					
SUN 2010					
TAN 2010					
Uyar 2019	ZBI	SF-36 ²	BDI	BAI	NPI-D
Wang 2010b					
Wang 2012e		WHO-QoL-Bref			
Wang 2014a					
Yang 2017a					
Zarepour 2020				SQ	
ZHAO 2010					
Number of studies	989	4	4	3	3
Total sample size	482	228	223	178	166
(Control)	507	115	108	90	82
(Intervention)	989	113	115	88	84

Table 2. Summary of care recipient outcome measures³

	Cognitive function	Quality of life	Activity level	Neuropsychiatric symptoms
Amit 2008				
Chen 2020	MoCA		BADL	BEHAVE-AD
Ghaffari 2019				
Govindakumari 2020		Unspecified		
Guerra 2011		DEMQOL		
Hinton 2020				
JIANG 2012	MMSE			
Lin 2017				
Lök 2017				
Mahdavi 2017				
Mercedeh 2014				
Orawan 2018				
Pahlavanzadeh 2010				
Pan 2019				
Salamizadeh 2017				
Shata 2017				
Söylemez 2016				
SU 2012	MMSE			

² Uyar (2019) only measured mental health and physical health quality of life scores, but not the general quality of life score. Thus, we were unable to include this study in the meta-analysis.

³ Please refer to the Abbreviations section for measurement methods used.

SUN 2010		SF-36		
TAN 2010	MMSE			
Uyar 2019		QoL-AD		NPI-S
Wang 2010b	MMSE		IADL	
Wang 2012e				
Wang 2014a	MMSE		IADL	
Yang 2017a	MMSE			
Zarepour 2020				
ZHAO 2010		SF-36		
Number of studies	7	5	3	2
Total sample size	639	310	156	142
(Control)	319	155	79	72
(Intervention)	320	155	77	70

Risk of bias

Risk of bias assessment was performed as guided with RoB2 tool, the revised Cochrane risk-of-bias tool for randomized trials,¹⁷ and visualised with the R package “robvis”.¹⁸ Six included studies were assessed by two reviewers and the remaining one was assessed by all four reviewers, with the disagreement settled through group consensus. The risk of bias of the 22 previous studies was obtained from the records of the 2019 meta-analysis. Thus, the risk of bias is summarised in Table 3 and Figure 2, with the details can be found in Appendix 3.

Commented [MOU3]: Risk of bias in individual studies: Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information was used in data synthesis
 Risk of bias across studies: Specify any assessment of risk of bias that may affect the cumulative evidence (such as publication bias, selective reporting within studies)

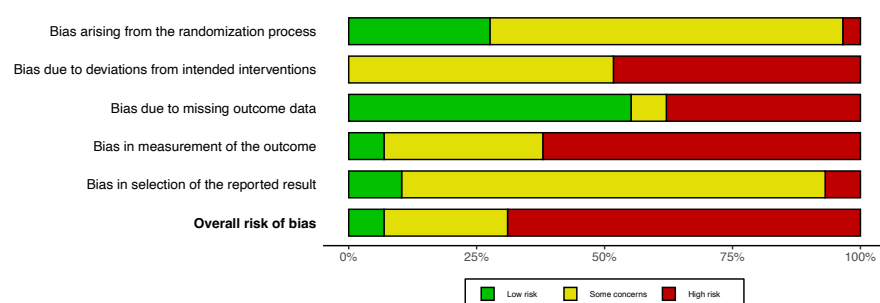


Figure 2. Risk of bias summary

Table 3. Risk of bias summary

	Risk of bias domains					Overall
	D1	D2	D3	D4	D5	
Study						
Amit 2008	+	-	+	-	-	-
Chen 2020	+	-	+	+	+	-
Ghaffari 2019	-	-	+	-	-	-
Govindakumari 2020	-	-	-	-	-	-
Guerra 2011	+	-	+	-	-	-
Hinton 2020	+	-	+	-	+	+
JIANG 2012	-	-	+	-	-	-
Lin 2017	-	-	+	-	-	-
Lök 2017	-	-	-	-	-	-
Mahdavi 2017	-	-	-	-	-	-
Mercedeh 2014	-	-	-	-	-	-
Orawan 2018	-	-	+	-	-	-
Pahlavanzadeh 2010	-	-	-	-	-	-
Pan 2019	+	-	-	-	-	-
Salamizadeh 2017	-	-	-	-	-	-
Shata 2017	+	-	-	-	-	-
Söylemez 2016	-	-	-	-	-	-
SU 2012	-	-	+	-	-	-
SUN 2010	-	-	+	-	-	-
TAN 2010	-	-	+	-	-	-
Uyar 2019	+	-	+	-	-	-
Wang 2010b	-	-	+	-	-	-
Wang 2012e	-	-	-	-	-	-
Wang 2014a	-	-	+	-	-	-
Wang 2017c	-	-	-	-	-	-
Wang 2017e	-	-	-	-	-	-
Yang 2017a	-	-	-	-	-	-
Zarepour 2020	+	-	+	+	+	+
ZHAO 2010	-	-	+	-	-	-

Domains:
D1: Bias arising from the randomization process.
D2: Bias due to deviations from intended intervention.
D3: Bias due to missing outcome data.
D4: Bias in measurement of the outcome.
D5: Bias in selection of the reported result.

Judgement
- High
- Some concerns
+ Low

Qualitative synthesis

Of 27 trials, 12 were based in China, 6 in Iran, 3 in Turkey, 2 in India, and 1 each in Egypt, Peru, Thailand and Vietnam, with the date of publication ranging from 2008 to 2020 (Figure 3). The trials were mostly randomised on an individual level, with only two studies randomised subjects based on the community. The subjects of study were dementia patients and their caregivers in all included trials. Over 70% percentage of caregivers were females, suggesting a significant gender difference consistent across studies.¹⁹

Commented [MOU4]: *Summary measures:* State the principal summary measures (such as risk ratio, mean difference)
Synthesis of results: Describe the methods of combining results of studies, including measures of consistency (such as I²) for each meta-analysis

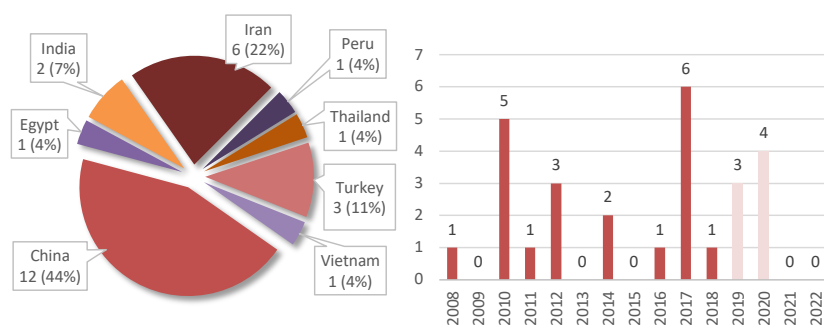


Figure 3. Selected characteristics of included studies

There was significant diversity in the content and duration of intervention. We categorised the intervention approaches into 1) structured training programmes, which typically involved training sessions covering prepared topics on a regular basis; 2) advisory team, which typically involved an advisory team providing suggestions and training on a request basis; 3) comprehensive support, which typically involved more active forms of caregiver support, e.g. regular phone calls and home visits, in which interventionists provided tailored suggestions and training. 14 studies took the form of structured training programme. Most training sessions were held weekly, with the median intervention period ranging from 4 weeks to 24 weeks. The content of training was typically skills and knowledge for dementia care and caregiver burden. Advisory teams without training sessions were provided in two studies, which lasted 8 weeks and 6 months respectively, while advisory teams during the structured training programme lasted 4 weeks and 5 months respectively. 10 studies provided proactive support for caregivers through comprehensive support schemes, with the median duration of a year.

18 trials measured caregiver outcomes, of which 12 measured perceived caregiver burden, 7 measured quality of life, 4 measured depression, 3 measured neuropsychiatric symptoms and 3 measured caregiver anxiety. 12 measured care recipient outcomes, including cognitive function measured in 7 studies, quality of life measured in 5 studies, activity level measured in 3 studies and neuropsychiatric symptoms measured in 2 studies

Statistical analysis

Commented [MOU5]: Please read
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6890471/>

Statistical analysis was performed with R. Random-effect meta-analyses were implemented with the R package 'meta' and 'metafor'.^{43–45} Change scores and its SMD were calculated as per Morris (2008).⁴⁶ They were used for the analyses as per the Cochrane Handbook. Sensitivity analyses of different measurement time points and measurement scales were conducted to examine the robustness of results.^{47,48} Subgroup analysis were done to identify the causes of heterogeneity and further explains the meta-analytic model.

DISCUSSION

Commented [MOU6]: The **discussion** section should have the following sub-headings:

- Statement of principal findings
- Strengths and weaknesses of the study
- Strengths and weaknesses in relation to other studies
- Meaning of the study: possible explanations and implications for clinicians and policymakers
- Unanswered questions and future research

Across these sub-headings, the discussion section should include information on all of the following elements (These are taken directly from the PRISMA statement and checklist):

Summary of evidence: Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (such as health care providers, users, and policy makers)

Limitations: Discuss limitations at study and outcome level (such as risk of bias), and at review level (such as incomplete retrieval of identified research, reporting bias)

CONCLUSIONS

Commented [MOU7]: The **conclusions** section should provide a brief overview of the general interpretation of the results and implications for research and practice.

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APPENDIX 1: SEARCH STRATEGY

	Condition	Ovid Medicine	Ovid Embase	Ovid Global Health	Ovid PsycINFO	EBSCO CINAHL	Total
1	Include dementia-related terms	427238	723805	34331	185529	159287	
2	Include therapy-related terms	13930216	18103622	2046951	2371714	3652534	
3	Include RCT-related terms	5552799	5229127	632419	755859	982366	
4	Include LMIC-related terms	4046862	4706713	1683278	504266	705117	
5	Satisfy Conditions 1, 2, 3 & 4	7917	7967	1911	2359	1917	
6	Nonhuman experiments		31095663				
7	Human experiments		24104793				
8	Condition 6 results excluding Condition 7 results	5049406	6992081		379966	210517	
9	Condition 5 results excluding Condition 8 results	6646	7556		2301	1819	
10	Publication type is Editorial or Letter	2097191	1977082				
11	Condition 9 results excluding Condition 10 results	6594	7540				
12	Year of publication limited to 2019-2020	1156	1247	382	404	448	
	Total results:	1156	1247	382	404	448	
	Duplicates:	(0)	(357)	(143)	(212)	(272)	(984)
	Results for screening:	1156	890	239	192	176	2653

Searches updated and ran by Andra Fry on 21/09/2022.

Appendix 1.1 Dementia-related terms

Dementia/ or Pick Disease of the Brain/ or Huntington Disease/ or Lewy Body Disease/
or Cognitive Dysfunction/ or (dement* or amentia* or (major adj3 cognit* adj3 disorder*)
or alzheimer* or alzeimer* or (cortical adj4 sclerosis) or ((encephalopath* or cogniti*
or neurocogniti*) adj4 (aids or acquired immun?deficiency syndrome* or acquired
immun? deficiency syndrome* or hiv or human immun?deficiency virus* or human
immun? deficiency virus*)) or pick* disease* or (lobar adj3 atroph* adj3 brain) or
Huntington* disease* or Huntington* chorea or (Lewy bod* adj3 disease*) or (cerebr*
adj3 deteriorat*) or (cerebr* adj3 insufficien*) or ((frontotemporal or fronto temporal or
corticobasal or cortico basal or frontal lobe*) adj4 (degenerati* or dysfunction*)) or
((cognit* or memory or cerebr*) adj3 (declin* or impair* or los* or deteriorat* or
degenerat* or insufficien*)) or MCI or (mild adj2 cognit* impair*).ti,ab,kf,kw.

Appendix 1.2 Dementia therapy-related terms

Cognitive Therapy/ or exp Drug Therapy/ or exp Cholinesterase Inhibitors/ or exp Antipsychotic Agents/ or exp Serotonin Uptake Inhibitors/ or exp Benzodiazepines/ or exp "Hypnotics and Sedatives"/ or exp Exercise/ or exp Exercise Therapy/ or exp Psychotherapy/ or exp Counseling/ or Psychosocial Support Systems/ or exp Complementary Therapies/ or Phototherapy/ or exp Advance Care Planning/ or Case Management/ or Caregivers/ed, px or Self-Help Groups/ or exp Social Support/ or Diagnosis, Computer-Assisted/ or Telemedicine/ or exp Computers, Handheld/ or (intervention* or therap* or treatment* or program* or manage* or prevent* or diagnos* or polic* or (cognit* adj3 (therap* or training or rehab*)) or cognit* psycho therap* or cognit* psychotherap* or drug* or medicine* or pharmacotherap* or pharmaco* therap* or cholinesterase agent* or cholinesterase inhibitor* or (tranquili* adj3 (agent* or drug*)) or antipsychotic* or (neuroleptic adj3 (agent* or drug*)) or serotonin uptake inhibitor* or serotonin reuptake inhibitor* or ssri* or benzodiazepine* or (sedative adj3 (effect* or agent*)) or memantine or donepezil or rivastigmine or galantamine or souvenaid or risperidone or haloperidol or olanzapine or quetiapine or citalopram or dextromethorphan or carbamazepine or mirtazapine or sertraline or moclobemide or trazodone or melatonin or ramelteon or methylphenidate or exercis* or physical activit* or physical training or aerobic* or arobic* or kinesi?therap* or kinesi? therap* or (social adj3 (activit* or engag* or stimul*)) or psycholog* therap* or psychotherap* or ((behavio?r* or conditioning) adj3 therap*) or counsel?ing or ((psychosocial or psycho social) adj3 (support or interven* or care)) or ((alternative or compl?ment* or traditional) adj3 (medicine* or therap*)) or acupunct* or (herb* adj3 (tea* or remedy or remedies or medicine*)) or ginkgo or homeopath* or ((music or art or aroma or light or photo or pet or pets) adj3 therap*) or aromatherap* or phototherap* or massage* or (mind adj3 body) or (advance? adj3 (care or medical or healthcare) adj3 plan*) or (decision* adj3 (aid* or support)) or (communicati* adj3 skill* adj3 training) or (dementia care adj3 map*) or ((person* or patient*) adj3 cent* adj3 care) or "Resources for Enhancing Alzheimer's Caregiver Health" or "Strategies for Relatives" or ((caregiver* or carer*) adj3 educat*) or telemedicine or tele medicine or (smart adj2 (phone* or device* or tablet*)) or smartphone* or cognitive aid* or reminder* or robot*).ti,ab,kf,kw.

Appendix 1.3 Dementia therapy-related terms

Developing Countries or (Africa* or Asia* or Caribbean* or West Indies or South America* or Latin America* or Central America* or Afghan* or Albania* or Algeria* or Angola* or Anguilla* or Antigua* or Barbuda* or Argentin* or Armenia* or Azerbaijan* or Azeri or Bangladesh* or Barbad* or Benin* or Byelarus or Byelorussian or Belarus* or Belorussia* or Belize* or Bhutan* or Bolivia* or Bosnia* or Herzegovin* or Hercegovin* or Botswana or Botsuana or Motswana or Batswana or Brasil* or Brazil* or Burkina Faso or Burkina Fasso or Burkina* or Burundi* or Urundi* or Cambodia* or Cameroon* or Cameron* or Cape Verd* or Cabo Verde or Central African Republic or Chad* or Tchad* or Chile* or China or Chinese or Colombia* or Columbia* or Comoros or Comoro Islands or Comores or Comoran or Mayotte or Congo* or Costa Rica* or Cote d'Ivoire or Ivory Coast or Ivorian* or Cook Islands or Cuba* or Croat* or Djibouti* or Dominica* or East Timor or East Timur or Timor Leste or Timorese or Ecuador* or Equador* or Egypt* or El Salvador or Salvadoran or Eritrea* or Ethiopia* or Fiji* or Gabon* or Gambia* or Gaza or Georgia Republic or Georgian or Abkhazia* or Abchasia* or South Ossetia* or Ghana* or Grenada or Grenadian or Guatemala* or Guinea* or Guinea Bissau or Guian* or Guyana or Haiti* or Hondura* or India or Indian or Indonesia* or Iran* or Iraq* or Jamaica* or Jordan* or Kazakhstan* or Kazakh or Kenya* or Kiribati or Korea* or Kosovo or Kosova* or Kyrgyzstan or Kirghizia or Kyrgyz or Kirghiz or Kirgizstan or Lao PDR or Laos or Laotian or Lebanon or Lebanese or Lesotho or Mosotho or Basotho or Liberia* or Libya* or Macedonia* or FYROM or Madagasca* or Malagasy or Malaysia* or Malaya* or Malay or Sabah or Sarawak or Malawi* or Maldives or Maldivan or Mali or Malian or Marshall Islands or Marshallese or Mauritania* or Mauriti* or Agalega Islands or Mexico or Mexican or Micronesia* or Middle East* or Moldova* or Moldovia* or Transnistria* or Mongolia* or Montenegro* or Montserrat* or Morocc* or Mozambique or Mozambican or Myanmar* or Myanma or Burma or Burmese or Namibia* or Nauru* or Niue or Nepal* or Nicaragua* or Niger or Nigerien or Nigeria* or Oman* or Pakistan* or Palau* or Palestine or Palestinian or Panama or Panamanian or Paraguay* or Papua New Guinea* or Peru or Peruvian or Philippines or Philipines or Phillipines or Phillippines or Filipino or Philipino or Philippino or Phillipino or Phillippino or Rwanda* or Ruanda* or Saint Helen* or St

Helen* or Saint Kitts or St Kitts or Kittian or Nevis* or Saint Lucia* or St Lucia* or Saint Vincent or St Vincent or Vicentian or Grenadines or Samoa* or Sao Tome* or Senegal* or Serbia* or Seychell* or Sierra Leone* or Sri Lanka* or Ceylon or Solomon Island* or Somali* or South Africa* or Sudan or Sudanese or Surinam* or Swaziland or Swazi or Eswatini or Syria or Syrian or Tajikistan or Tadjhikistan or Tadjikistan or Tajik or Tadjhik or Tanzania* or Thailand or Thai or Togo or Togolese or Tonga* or Tunisia* or Tokelau or Trinidad* or Tobago* or Turkey or Turkish or Turks or Turkmenistan or Turkmen or Tuvalu* or Uganda* or Ukraine or Ukrainian or Uruguay* or Uzbekistan* or Uzbek or Vanuatu or Venezuela* or Vietnam* or Viet Nam or Wallis Futuna or West Bank or Yemen* or Zambia* or Zimbabwe*).hw,kf,kw,ti,ab,cp. or (((developing or less* developed or under developed or underdeveloped or middle income or low* income or underserved or under served or deprived or poor*) adj (countr* or nation? or population? or world)) or ((developing or less* developed or under developed or underdeveloped or middle income or low* income) adj (economy or economies)) or (low* adj (gdp or gnp or gross domestic or gross national)) or (low adj3 middle adj3 countr*) or Imic or Imics or third world or lami countr* or transitional countr*).ti,ab.

APPENDIX 2 SELECTED CHARACTERISTICS OF INCLUDED STUDIES

Study identifier	Country	Descriptive name of the intervention	Intervention approach(es)	Intervention frequency	Randomisation unit	Sample size (At baseline)		Intervention duration	Follow-up duration (from baseline)
						Control	Intervention		
Hinton 2020 ²⁰	Vietnam	REACH VN	Structured training programme	Weekly	Community	26	25	3 months	3 months
Govindakumari 2020 ²¹	India	Home-based cognitive training program	Structured training programme	Twice a week	Individual	157	157	1 month	1 month
Chen 2020 ²²	China	Extended Nursing	Advisory and support team	N/A	Individual	51	51	8 weeks	1 year
Zarepour 2020 ²³	Iran	Educational Intervention	Structured training programme; advisory and support team	7 sessions in 4 weeks	Individual	35	35	4 weeks	4weeks + 1 month 4weeks + 3 months
Pan 2019 ²⁴	China	Nurse-led CBI on depressive symptoms for PwD caregivers	Structured training programme; advisory and support team	Monthly	Individual	56	56	5 months	7 months
Ghaffari 2019 ²⁵	Iran	Resilience Education	Structured training programme	Weekly	Individual	27	27	8 weeks	8 weeks
Uyar 2019 ¹⁵	Turkey	Dementia care and support program	Comprehensive support	N/A	Individual	28	26	16 weeks	16 weeks
Amit 2008 ²⁶	India	Home Care Program for supporting caregivers of PwD	Advisory and support team	N/A	Individual	40	41	6 months	3 months 6 months
Guerra 2011 ¹⁶	Peru	'Helping Carer to Care' training	Structured training programme	Weekly	Individual	29	29	5 weeks	6 months
JIANG 2012 ²⁷	China	Whole course rehabilitation nursing intervention	Comprehensive support	N/A	Individual	48	48	1 year	1 year
Liu 2017 ^{28 4}	China	Family visits of community nurses	Comprehensive support	N/A	Individual	39	39	3 months	3 months
Lök 2017 ²⁹	Turkey	"First You Should Get Stronger" Caregiving Program	Structured training programme	Weekly	Individual	20	20	7 weeks	7 weeks
Mahdavi 2017 ³⁰	Iran	Spiritual Group Therapy	Structured training programme	Weekly	Individual	35	32 ⁵ / 33 ⁶	5 weeks	5 weeks

⁴ The study identifier in the 2019 meta-analysis records was "Lin 2017", which proved to be a typo during full-text screening.

⁵ Control group 1: Group sessions without spiritual therapy

⁶ Intervention group: Spiritual group therapy

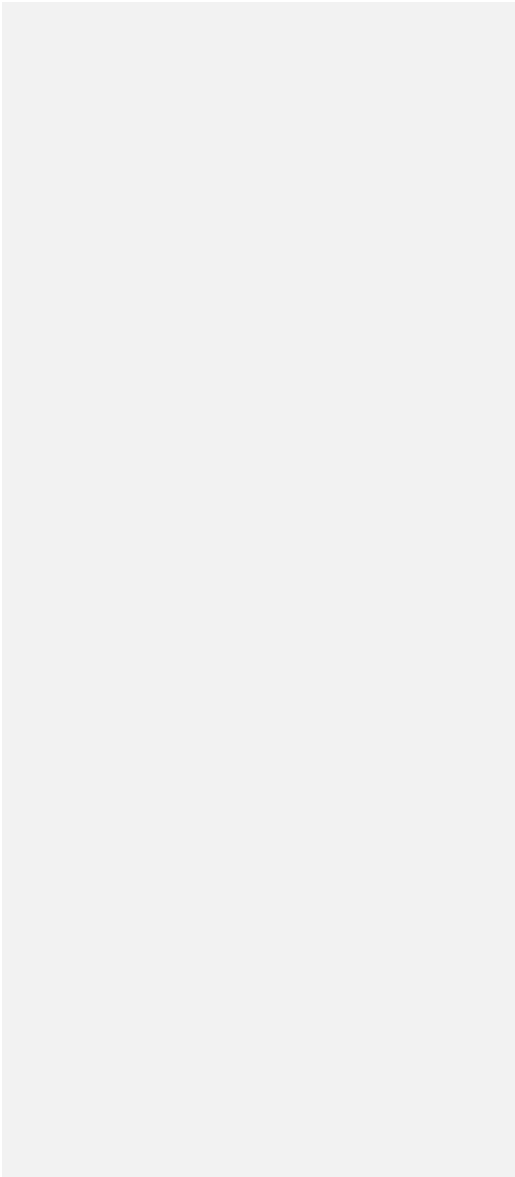
Mercedeh 2014 ³¹	Iran	Mindfulness based cognitive therapy	Structured training programme	Weekly	Individual ⁷	10	10	8 weeks	8 weeks 8 weeks + 2 months
Orawan 2018 ¹²	Thailand	Enhancing Positive Aspects of Caregiving Program	Structured training programme	Weekly	Individual	40	40	7 weeks	8 weeks 12 weeks 20 weeks
Pahlavanzadeh 2010 ³²	Iran	Family education program	Structured training programme	Weekly	Individual	30	30	5 weeks	5 weeks 5 weeks + 1 month
Salamizadeh 2017 ¹³	Iran	Spiritual Care Education	Structured training programme	Weekly	Individual ⁸	28	26	5 weeks	5 weeks
Shata 2017 ³³	Egypt	A multi-component psychosocial intervention program	Structured training programme	Weekly	Individual	59	55	8 weeks	8 weeks 8 weeks + 3 months
Söylemez 2016 ³⁴	Turkey	Progressively Lowered Stress Threshold-based intervention	Comprehensive support	Bi-weekly	Individual	35	35	3 months	5 months + 2 weeks ⁹
SU 2012 ³⁵	China	Comprehensive nursing intervention	Comprehensive support	N/A	Individual	20	20	14 days	14 days + 6 months
SUN 2010 ³⁶	China	Nursing intervention	Comprehensive support	N/A	Individual	19	19	1 year	1 year
TAN 2010 ³⁷	China	Early family intervention	Comprehensive support	N/A	Individual	26	26	1 year	1 year
Wang 2010b ³⁸	China	Community nursing intervention	Comprehensive support	N/A	Individual	40	40	1 year	1 year
Wang 2012e ³⁹	China	Mutual support group	Structured training programme	Bi-weekly	Individual	39	39	24 weeks	24 weeks 24 weeks + 1 month
Wang 2014a ⁴⁰	China	Home care intervention	Comprehensive support	N/A	Individual	64	64	1 year	6 months 12 months
Yang 2017a ⁴¹	China	Family medical intervention model	Comprehensive support	N/A	Community	70	71	1 year	6 months 1 year
ZHAO 2010 ⁴²	China	Family nursing intervention	Comprehensive support	N/A	Individual	25	25	1 year	1 year

⁷ The authors claimed that the study was a quasi-experiment, but the study randomised its participants.

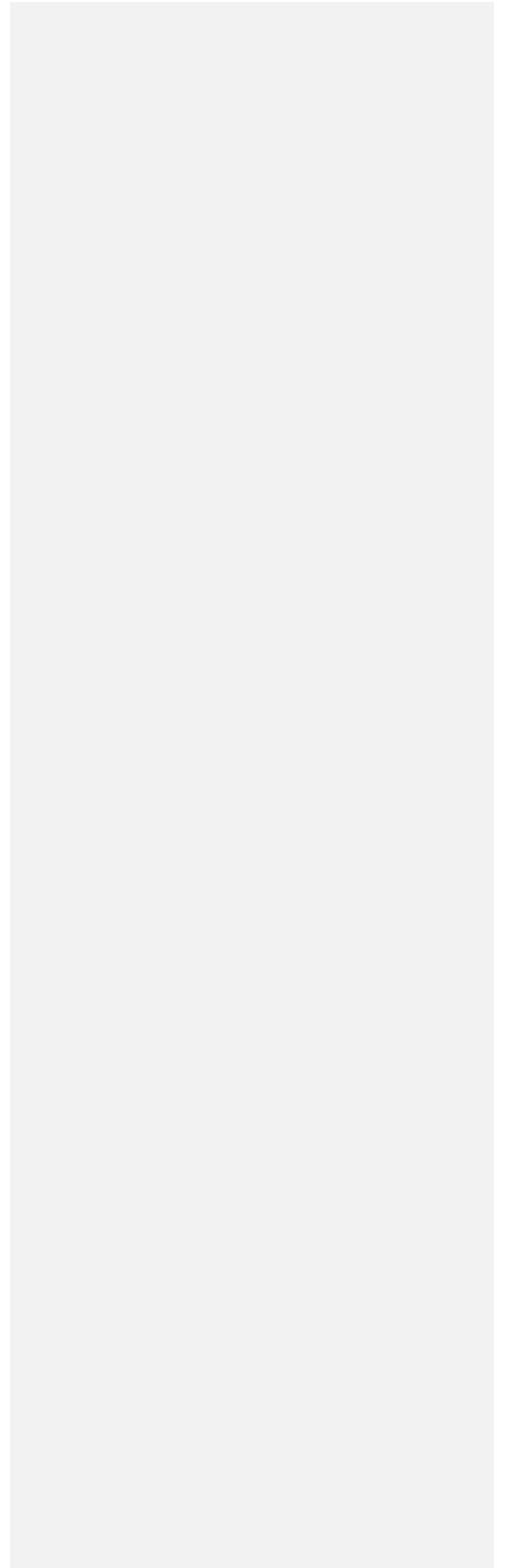
⁸ The authors claimed that the study was a quasi-experiment, but the study randomised its participants.

⁹ The second home visits took place two weeks after the first visit. The third home visit took place 2 months after the second visit. The measurement occurred 3 months after the third visit.

APPENDIX 3: RISK OF BIAS OF 7 STUDIES FROM THE 2022 SEARCH



APPENDIX 4: DATA PROCESSING PROCEDURES



HP407

Candidate No. 51600

APPENDIX 5: DATA ANALYSIS METHODS

HP407

Candidate No. 51600